

Ripasso derivate

Fonte esercizi:

<https://www.math-exercises.com/limits-derivatives-integrals/derivative-of-a-function>

Prima Tabella

a)	$y = x^2 + 2x + 1$	j)	$y = x + \sqrt{x} + \sqrt[3]{x} + \sqrt[5]{x}$	s)	$y = \sin x + \cos x + \tan x$
b)	$y = 4x^3 - 3x^2 + 2x - 1$	k)	$y = \sqrt[3]{x^8} - \sqrt[4]{x^7} + \sqrt[5]{x^6}$	t)	$y = \log x - \ln x + \log_5 x$
c)	$y = \frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2$	l)	$y = \frac{1}{\sqrt[3]{2x^4}} - \frac{1}{\sqrt[4]{2x^3}}$	u)	$y = 3 \arcsin x - 2 \arctan x$
d)	$y = -\frac{5x^8}{9} - \frac{8x^7}{13} - \frac{9x^6}{16}$	m)	$y = \frac{\sqrt[5]{4x}}{5} + 4\sqrt[4]{\frac{1}{x^3}}$	v)	$y = 5 \cot x + 8 \arccos x$
e)	$y = (3x - 5)^3$	n)	$y = \sqrt{x^5} - 4\sqrt{x^9}$	w)	$y = \operatorname{arccot} x - 2 \cot x$
f)	$y = (\sqrt{x} - 1)^2 - (x^2 + 1)^4$	o)	$y = \sqrt{x\sqrt{x}} + \sqrt[3]{x^2\sqrt{x^3}}$	x)	$y = x - \ln x + 1$
g)	$y = x^{11} - x^9 + x^7 - x^5$	p)	$y = \sqrt{x^3\sqrt{x^5\sqrt{x^7}}}$	y)	$y = 2^x - 3e^x - 4^x$
h)	$y = x^{-5} + x^{-7} + x^{-9} - 11$	q)	$y = \frac{5x^{-3} \cdot \sqrt{x^4} \cdot \sqrt[3]{x^5}}{8x^9 \cdot \sqrt[5]{x^{-8}} \cdot \sqrt[7]{x^{11}} \cdot \sqrt[9]{x}}$	z)	$y = 5 \times 9^x - 4 \times 5^x + \frac{7^x}{\ln 7}$
i)	$y = \frac{8}{x^8} - \frac{6}{x^6} + \frac{4}{x^4} - \frac{2}{x^2}$	r)	$y = \sqrt{\frac{\sqrt[5]{x^7}}{6x^3}} + \frac{\sqrt[3]{x^{-8}} \cdot \sqrt{x^9}}{\sqrt{4x\sqrt{x^5}}}$	Z)	$y = -6e^x + 5^x - 5x + \frac{x}{5}$

Risoluzioni

1.

$$f(x) = x^2 + 2x + 1$$

Sol. $f'(x) = 2x + 2$

2.

$$f(x) = 4x^3 - 3x^2 + 2x - 1$$

Sol. $f'(x) = 12x^2 - 6x + 2$

3.

$$f(x) = \frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2$$

Sol. $f'(x) = x^3 + x^2 + x$

4.

$$f(x) = -\frac{5}{9}x^8 - \frac{8}{13}x^7 - \frac{9}{16}x^6$$

Sol. $f'(x) = -\frac{40}{9}x^7 - \frac{56}{13}x^6 - \frac{54}{16}x^5$

5.

$$f(x) = (3x - 5)^5$$

Sol. $f'(x) = 5(3x - 5)^4 \cdot 3$
 $= 15(3x - 5)^4$

6.

$$f(x) = (\sqrt{x} - 1)^2 - (x^2 + 1)^4$$

Sol. $f'(x) = 2(\sqrt{x} - 1) \cdot \frac{1}{2\sqrt{x}} - 4(x^2 + 1)^3 \cdot 2x$
 $= \frac{\sqrt{x} - 1}{\sqrt{x}} - 8x(x^2 + 1)^3$

7.

$$f(x) = x^{11} - x^9 + x^7 - x^5$$

Sol. $f'(x) = 11x^{10} - 9x^8 + 7x^6 - 5x^4$

8.

$$f(x) = x^{-5} + x^{-7} + x^{-9} + -11$$

Sol. $f'(x) = -5x^{-6} - 7x^{-8} - 9x^{-10}$

9.

$$f(x) = \frac{8}{x^8} - \frac{6}{x^6} + \frac{4}{x^4} - \frac{2}{x^2}$$

$$= 8x^{-8} - 6x^{-6} + 4x^{-4} - 2x^{-2}$$

$$\text{Sol. } f'(x) = -64x^{-9} + 36x^{-7} - 16x^{-5} + 4x^{-3}$$

10.

$$f(x) = x + \sqrt{x} + \sqrt[3]{x} + \sqrt[5]{x}$$

$$= x + x^{1/2} + x^{1/3} + x^{1/5}$$

$$\text{Sol. } f'(x) = 1 + \frac{1}{2\sqrt{x}} + \frac{1}{3\sqrt[3]{x^2}} + \frac{1}{5\sqrt[5]{x^4}}$$

11.

$$f(x) = \sqrt[3]{x^8} - \sqrt[4]{x^7} + \sqrt[5]{x^6}$$

$$= x^{8/3} - x^{7/4} + x^{6/5}$$

$$\text{Sol. } f'(x) = \frac{8}{3}x^{5/3} - \frac{7}{4}x^{3/4} + \frac{6}{5}x^{1/5}$$

12.

$$f(x) = \frac{1}{\sqrt[3]{2x^4}} + \frac{1}{\sqrt[4]{2x^3}}$$

$$= (2x^4)^{-\frac{1}{3}} + (2x^3)^{-\frac{1}{4}}$$

$$\text{Sol. } f'(x) = -\frac{1}{3}(2x^4)^{-\frac{4}{3}} \cdot 8x^3 - \frac{1}{4}(2x^3)^{-\frac{5}{4}} \cdot 6x^2$$

$$= -\frac{8x^3}{3\sqrt[3]{(2x^4)^4}} - \frac{6x^2}{4\sqrt[4]{(2x^3)^5}}$$

13.

$$f(x) = \frac{\sqrt[5]{4x}}{5} + \sqrt[4]{\frac{1}{x^3}}$$

$$= \frac{1}{5}(4x)^{1/5} + x^{-3/4}$$

$$\text{Sol. } f'(x) = \frac{1}{25}(4x)^{-4/5} \cdot 4 - \frac{3}{4}x^{-7/4}$$

$$= \frac{4}{25\sqrt[5]{(4x)^4}} - \frac{3}{4\sqrt[4]{x^7}}$$

14.

$$\begin{aligned}
 f(x) &= \sqrt{x^5} - \sqrt[4]{x^9} \\
 &= x^{5/2} - x^{9/4}
 \end{aligned}$$

Sol.

$$\begin{aligned}
 f'(x) &= \frac{5}{2}x^{3/2} - \frac{9}{4}x^{5/4} \\
 &= \frac{5\sqrt{x^5}}{2} - \frac{9\sqrt[4]{x^5}}{4}
 \end{aligned}$$

15.

$$\begin{aligned}
 f(x) &= \sqrt{x\sqrt{x}} + \sqrt[3]{x^2\sqrt{x^3}} \\
 &= (xx^{1/2})^{1/2} + (x^2x^{3/2})^{1/3} \\
 &= (x^{3/2})^{1/2} + (x^{7/2})^{1/3} \\
 &= x^{3/4} + x^{7/6}
 \end{aligned}$$

Sol.

$$\begin{aligned}
 f'(x) &= \frac{3}{4}x^{-1/4} + \frac{7}{6}x^{1/6} \\
 &= \frac{3}{4\sqrt[4]{x}} + \frac{7\sqrt[6]{x}}{6}
 \end{aligned}$$

16.

$$\begin{aligned}
 f(x) &= \sqrt{\frac{\sqrt[5]{x^7}}{6x^3}} + \frac{\sqrt[3]{x^{-8}\sqrt{x^9}}}{\sqrt{4x\sqrt{x^5}}} \\
 &= (x^{7/5} \cdot 6x^{-3})^{1/2} + (x^{-8} \cdot x^{9/2})^{1/3} \cdot (4x \cdot x^{5/2})^{-1/2} \\
 &= \left(\frac{1}{6}x^{-8/5}\right)^{1/2} + (x^{-7/2})^{1/3}(4x^{7/2})^{-1/2} \\
 &= \frac{1}{\sqrt{6}}x^{-8/10} + x^{-7/6}x^{-7/4}4^{-1/2} \\
 &= \frac{1}{\sqrt{6}}x^{-4/5} + \frac{x^{-14/12-21/12}}{2} \\
 &= \frac{1}{\sqrt{6}}x^{-4/5} + \frac{1}{2}x^{-35/12}
 \end{aligned}$$

Sol.

$$f'(x) = -\frac{4}{5\sqrt{6}}x^{-9/5} - \frac{35}{24}x^{-47/12}$$

17.

$$f(x) = \log x - \ln x + \log_5 x$$

$$\text{Sol. } f'(x) = \frac{1}{x \ln 10} - \frac{1}{x} + \frac{1}{x \ln 5}$$

18.

$$f(x) = x - \ln x + 1$$

$$\text{Sol. } f'(x) = 1 - \frac{1}{x}$$

19.

$$f(x) = 2^x - 3e^x - 4^x$$

$$\text{Sol. } f'(x) = 2^x \ln 2 - 3e^x - 4^x \ln 4$$

20.

$$f(x) = 5 \times 9^x - 4 \times 5^x + \frac{7^x}{\ln 7}$$

$$\begin{aligned} \text{Sol. } f'(x) &= 5 \times 9^x \ln 9 - 4 \times 5^x \ln 5 + \frac{7^x \ln 7}{\ln 7} \\ &= 5 \times 9^x \ln 9 - 4 \times 5^x \ln 5 + 7^x \end{aligned}$$
